4.2.5. Titanic Dataset Analysis and Data Cleaning

20:38

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarke

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Code:

import pandas as pd

```
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())

# 2. Display the last 5 rows of the dataset
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())

# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the ‘Age’ column with the median age
median_age = data['Age'].median()
```

```
data['Age'].fillna(median_age, inplace=True)

# 8. Fill missing values in the ‘Embarked’ column with the mode
mode_embarked = data['Embarked'].mode() [0]
data['Embarked'].fillna(mode_embarked, inplace=True)

# 9. Drop the ‘Cabin’ column due to many missing values
data.drop('Cabin', axis=1, inplace=True)

# 10. Create a new column 'FamilySize’ by adding ‘SibSp' and ‘Parch'
data['FamilySize']=data['SibSp']+data['Parch']
```

Average time
1.171 s
1171.00 ms

Maximum time
1.171 s
1171.00 ms

1 out of 1 shown test case(s) passed

Test case 1 1171 ms Debug

Expected output	Actual output
`   PassengerId  Survived  Pclass  ...     Fare Cabin  E mbarked`	`   PassengerId  Survived  Pclass  ...     Fare Cabin  Embarked`
`0            1         0       3  ...   7.2500   NaN         S`	`0            1         0       3  ...   7.2500   NaN         S`
`1            2         1       1  ...  71.2833   C85         C`	`1            2         1       1  ...  71.2833   C85         C`
`2            3         1       3  ...   7.9250   NaN         S`	`2            3         1       3  ...   7.9250   NaN         S`
`3            4         1       1  ...  53.1000  C123         S`	`3            4         1       1  ...  53.1000  C123         S`
`4            5         0       3  ...   8.0500   NaN         S`	`4            5         0       3  ...   8.0500   NaN         S`
`[5 rows x 12 columns]`	`[5 rows x 12 columns]`
`     PassengerId  Survived  Pclass  ...   Fare Cabin  E mbarked`	`     PassengerId  Survived  Pclass  ...   Fare Cabin  Embarked`
`886          887         0       2  ...  13.00   NaN         S`	`886          887         0       2  ...  13.00   NaN         S`
`887          888         1       1  ...  30.00   B42         S`	`887          888         1       1  ...  30.00   B42         S`
`888          889         0       3  ...  23.45   NaN         S`	`888          889         0       3  ...  23.45   NaN         S`
`889          890         1       1  ...  30.00  C148         C`	`889          890         1       1  ...  30.00  C148         C`
`890          891         0       3  ...   7.75   NaN         Q`	`890          891         0       3  ...   7.75   NaN         Q`
`[5 rows x 12 columns]`	`[5 rows x 12 columns]`
`(891, 12)`	`(891, 12)`
`<class 'pandas.core.frame.DataFrame'>`	`<class 'pandas.core.frame.DataFrame'>`
`RangeIndex: 891 entries, 0 to 890`	`RangeIndex: 891 entries, 0 to 890`